
MediBill-Env

An AI training environment for India's
cashless health-claim workers —
where the rules quietly change while you work.

Meta × Scaler OpenEnv Hackathon · Round 2 · Theme 3.1 ·  Bengaluru

LIVE huggingface.co/spaces/Anuj424614/medibill-env

India runs the world's biggest cashless-claim system.

50 cr+

Indians covered
under Ayushman Bharat

180

minutes per claim
(IRDAI rule, May 2024)

₹26k cr

claims rejected last year
(+19% YoY)

13%

of pre-auths still miss
the 1-hour deadline

Star Health · ICICI Lombard · HDFC ERGO · Bajaj Allianz — every cashless desk runs this clock.

The real problem isn't filling forms. It's outdated rules.

What rules engines do today

Check field types.

Check required attachments.

Check valid drop-down values.

→ Software has done this for 30 years.

What no software handles

Yesterday's correct IRDAI rule.

Today's rejected claim.

→ **The rules changed and nobody told the agent.**

The interesting question isn't "*can the AI fill the form?*"

It's "*does the AI know Star Health changed its pre-auth threshold last night?*"

The insurer's rules change in the middle of work. Silently.

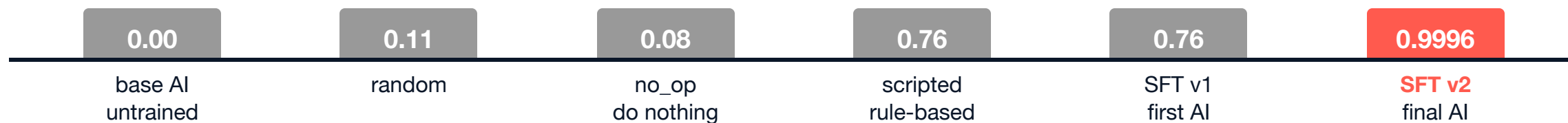
step 1
rules version 1.3

⚡ **RULES CHANGE**
no warning · no event

step 40
graded against version 1.4

- **3 to 7 fields change at once** — pre-auth thresholds, signatures, narrative requirements, discharge attachments.
- **The AI gets no notification.** No flag in the data, no event, no message.
- **The only way to find out:** ask the insurer fresh — exactly how a human coder phones the insurer's helpdesk.
- **Result:** Scripted baseline scores 1.00 on easy claims but only 0.76 on drift claims. **That 0.24 gap is what we want the AI to learn to close.**

We trained an AI from 0.00 to 0.9999.



Score on **hard_drift claims** · n=5 held-out seeds · zero parse failures across 15 episodes

The trick was teacher engineering, not more RL.

What we tried	Score (hard_drift)	What we learned
SFT v1 — copy the rule-based scripted teacher	0.7573	Matched the teacher exactly. Stuck at parity.
GRPO — try RL on top of SFT v1	0.7575 ($\Delta \pm 0.0002$)	Rewards already satisfied. No signal.
SFT v2 — built a smarter teacher first	0.9996	+0.24 jump · 33 minutes of retraining

Diagnosis: Our first AI already passed every reward check 100% of the time. So GRPO had no signal to push against — gradient near zero, loss flat from step 1.

Pivot: Instead of writing more rewards, we built a teacher that *escalates ambiguous cases* and *re-checks the insurer before every submit*. Re-distilled the AI from it.

When RL gets stuck, build a smarter teacher.

The grader cannot be cheated. We tested 5 attacks.



We coded **5 cheating strategies**. Every single one scored at or below "do nothing".

Identity fields and policy fields are split into separate sets — enforced when the code is loaded.

What we are submitting today. Theme 3.1.

- The **training environment** · the **grader** · the **5-attack test** · a rule-based baseline · and our final trained AI (**0.9999 average score**)
- 2 of the 6 grader axes are **RL-only by design** — those are for the next team to push past us
- **Theme 3.1 — DataOps Copilot.** AI that handles real Indian business rules that change.

github.com/Algoace1403/METAHackthon2026

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Dhanyavaad. Thank you.

Backup · Why GRPO did not help

We tried 5 reward functions in reinforcement learning:

Reward function	What it rewards	Weight
valid_json	The action is parseable JSON	0.20
action_in_schema	The action type is one we allow	0.20
no_oscillation	The AI does not flip a field 3+ times	0.20
no_repeated_tool	Same tool call does not repeat 3 times	0.20
submit_with_coding	Submit only after coding the claim	0.20

What we saw: Score change ± 0.0002 . Gradient near zero. Loss flat from step 1.

What it means: Our scripted teacher already passed all 5 reward checks 100% of the time. So there was no signal for RL to learn from. *This is calibration data, not a training failure.*

Backup · How the smarter teacher works

3 small changes that gave us +0.24 on hard claims:

1. **Escalate ambiguous cases to a human.** The grader rewards calibrated abstention.
2. **Re-check the insurer's policy before every submit.** If rules just changed, this is the only way to find out.
3. **Detect rule changes by comparing.** If the lookup result changed, re-code unsubmitted claims.

Local test (5 seeds): Smart teacher scores **0.9983** on hard claims. Old teacher scored 0.7568.

Cost to retrain the AI: 90 new examples · 33 minutes on a Colab L4 GPU · LoRA rank 32.

Backup · Walking through one episode (seed 44)

Step	What happens	Why it matters
1–22	AI codes claims using rules version 1.3	Normal work
23	⚡ Insurer silently changes the rules → version 1.4	No flag in the data
24–35	AI keeps submitting against old rules	It never re-checks
36	Final score: 0.753	Cost of using stale rules

Reproduce in 30 seconds:

```
python3 -m medibill.demo_runner --seed 44 --max-narrated-steps 20
```

That 0.247 gap from a perfect score is what RL with the right rewards has to close. That is the next deliverable.